PROJECTION FROM A GEOMETRICALLY IRREDUCIBLE FACTOR INDUCES A BIJECTION ON IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the topological criterion; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-bijection-irreducible-components is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Scheme theoretic fibres are homeomorphic to set theoretic fibres. Schemes, Lemma schemes-lemma-fibre-topological, invoked at proof lines 1394-1396. Supplies the comparison between the two notions of fibre.
- A scheme over a field is universally open over the base. Morphisms, Lemma morphisms-lemma-scheme-over-field-universally-open, invoked at proof lines 1397-1398. Supplies openness of the projection.
- An open map with irreducible fibres induces a bijection on irreducible components. Topology, Lemma topology-lemma-irreducible-fibres-irreducible-components, invoked at proof lines 1399-1401. Supplies the bijection once the hypotheses are in hand.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01K1: sources/stacks-chow/tags.txt, source lines 2034–2034. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 037A: sources/stacks-chow/tags.txt, source lines 4193–4193. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X, Y be schemes over k. Assume X is geometrically irreducible over k. Then the projection morphism*

$$p : X \times_k Y \longrightarrow Y$$

induces a bijection between irreducible components.

Proof. First, note that the scheme theoretic fibres of p are irreducible, since they are base changes of the geometrically irreducible scheme X by field extensions. Moreover the scheme theoretic fibres are homeomorphic to the set theoretic fibres, see Schemes, Lemma 01K1. By Morphisms, Lemma 0383 the map p is open. Thus we may apply Topology, Lemma 037A to conclude. □